CAPSTONE 10

AI Technical Report and Documentation Generator

Case Study: Automatic Street Light System Using LDR

Prepared by: _____

Department: Electronics and Communication Engineering (ECE)

College: _____

Academic Year: _____

Project question

How can AI help engineering students produce clear, correct project reports, README files and presentations while maintaining academic integrity and verifying every technical statement against real project data?

Abstract

This capstone demonstrates a responsible AI-assisted workflow for preparing engineering documentation. An Automatic Street Light System using a Light Dependent Resistor (LDR) is used as the example ECE project. The system senses ambient light and switches a lamp or LED ON in darkness and OFF in sufficient light. AI tools are used to turn student-provided notes into an abstract, introduction, methodology, results section, conclusion, README, user guide and presentation outline. AI-generated statements are treated as drafts and checked against circuit connections, component datasheets, source code and measured observations. No experimental measurements are invented in this report; the results table is intentionally provided for the student to complete after testing.

1. Introduction

Engineering students often spend significant time formatting reports, writing explanations and preparing presentations. Manual documentation can lead to inconsistent wording, missing sections and avoidable formatting errors. Generative AI can accelerate drafting, but it can also produce incorrect component values, unsupported claims or fabricated results. This project establishes a workflow in which AI assists with language and structure while the student remains responsible for technical verification.

2. Problem statement and objectives

• Manual report preparation is repetitive and time-consuming.

• Students may struggle to explain technical work in a clear, consistent format.

• AI-generated technical claims and numerical results must be verified.

• Build a reusable workflow that converts genuine lab notes into structured documentation.

• Measure actual manual and AI-assisted preparation time and compare document quality.

3. Selected ECE case study: Automatic Street Light Using LDR

An LDR is a light-sensitive resistor whose resistance generally decreases as incident light increases. A threshold-sensing circuit or microcontroller reads the light level and controls a transistor/relay driver for the lamp. For a safe classroom prototype, use a low-voltage LED load. Do not connect mains voltage unless supervised by a qualified instructor and using appropriate isolation and protection.

4. Components and working principle

• LDR: senses ambient illumination.

• Fixed resistor or potentiometer: forms a voltage divider / adjusts threshold.

• Comparator or microcontroller: compares the sensed level with a chosen threshold.

• NPN transistor or suitable driver: switches the LED load; use a base/gate resistor as required by the design.

• LED and current-limiting resistor: represent the street lamp.

• Regulated low-voltage DC supply and connecting wires/breadboard.

Working: In bright conditions, the sensor voltage is on one side of the chosen threshold and the controller keeps the lamp OFF. In darkness, the sensor voltage crosses the threshold and the controller switches the lamp ON. The exact voltage polarity depends on the voltage-divider arrangement; confirm it from the actual circuit before documenting it.

5. Methodology

• Build or simulate the LDR sensing circuit using the actual circuit diagram.

• Record the exact component values, supply voltage, connections and software version.

• Test under bright light, partial shade and darkness.

• For each condition, record sensor reading/voltage and lamp state; repeat readings if possible.

• Save photographs, simulation screenshots and raw observations.

• Enter the real notes and measurements into the prompt templates.

• Generate draft sections, then verify every technical claim and number.

• Export the checked report, README, user guide and presentation.

6. Manual documentation process and time

The manual baseline must be timed by the student. Start the timer when writing begins and stop when the first complete draft is ready. Record editing and verification separately. Then repeat the same task using the AI-assisted workflow. Do not estimate or invent times.

Task	Manual time (min)	AI-assisted time (min)	Evidence/notes
Outline + section drafting	_____	_____	_____
README + user guide	_____	_____	_____
Presentation preparation	_____	_____	_____
Technical checking + editing	_____	_____	_____
Total	_____	_____	_____

Time saved (%) = ((Manual total – AI-assisted total) ÷ Manual total) × 100. Calculate only after both timings are recorded.

7. AI prompt design

Use role + task + context + source data + output format + constraints. Tell the model not to invent readings, component values, citations or test results. Ask it to mark missing information as [TO BE PROVIDED].

8. Prompt library

Abstract

Act as an ECE technical editor. Using only the project notes below, write a 150–200 word abstract covering problem, objective, method and verified outcome. Do not invent results. Mark missing facts [TO BE PROVIDED]. Notes: [PASTE NOTES].

Introduction

Write a student-friendly technical introduction for [PROJECT]. Explain background, problem, need and objectives. Use only supplied facts; flag claims that need a source.

Methodology

Convert these actual build steps and circuit details into numbered methodology steps. Preserve all component values and pin numbers exactly. Flag ambiguity; do not assume missing connections. Data: [PASTE].

Results

Create a results table from these raw observations only. Do not calculate or infer missing values. Clearly distinguish measured, calculated and expected values. Data: [PASTE].

Conclusion

Write a concise conclusion based only on the verified results below. State limitations and future improvements. Do not claim success beyond the evidence. Results: [PASTE].

README

Create a GitHub README with project overview, features, components, circuit, setup, operation, testing, limitations and credits. Use placeholders for unknown details.

Presentation

Create a 7-slide outline: title, problem/objectives, components, block diagram, methodology, measured results, conclusion/future work. Keep each slide concise and label unmeasured results as pending.

9. AI tools used

Suggested tool set (use at least four and record the versions/access dates actually used): ChatGPT for drafting and prompt iteration; Gemini or Claude for an independent clarity review; Perplexity for locating datasheets or reputable references; Canva or Gamma for slide layout; GitHub for version control. Wokwi or Tinkercad may be used for simulation where the circuit is supported. Tool names are suggestions, not evidence that a tool was used.

10. Verification procedure

• Check each component name, rating and pinout against its manufacturer datasheet.

• Compare every circuit connection with the actual schematic and physical build/simulation.

• Check each numerical value against the raw lab notebook or instrument reading.

• Re-run the circuit in each test condition and record the observed output.

• Check references exist and support the statements attributed to them.

• Review grammar, tone, plagiarism risk and copied material; cite sources and obtain permission for reused figures.

• Keep an error log with the AI claim, correction, evidence and date.

11. Results and observation table

Fill this table only after performing the experiment. Add the actual sensor voltage or ADC reading and the actual lamp state.

Test condition	Measured LDR/sensor value	Lamp state observed	Photo/log reference
Bright light	_____	_____	_____
Partial shade	_____	_____	_____
Darkness	_____	_____	_____

12. Before-AI and after-AI quality comparison

Rate each draft from 1 (poor) to 5 (excellent), using the same rubric. Keep evidence such as draft versions, timestamps, correction count and reviewer comments.

Criterion	Before AI (1–5)	After AI (1–5)	Evidence
Completeness	____	____	_____
Clarity	____	____	_____
Technical correctness	____	____	_____
Formatting consistency	____	____	_____
Citation quality	____	____	_____

13. Automation

The included Python script reads a plain-text lab-notes file and creates a structured Markdown report outline. It does not call an AI API or fabricate content. The student can paste the outline into an approved AI tool with the prompt library, then verify the result.

14. Error log

Record every detected hallucination, incorrect value, unsupported claim, citation issue or tone problem. Suggested columns: date, AI output, issue, correction, evidence/source, reviewer and status. A blank CSV template is included.

15. Academic integrity, copyright and limitations

• AI is an assistant, not the author of experimental work. The student must understand and defend every section.

• Never submit invented measurements, simulated results presented as physical measurements, or fabricated references.

• Disclose AI assistance according to the institution's policy.

• Cite datasheets, articles, images and code. Check licenses and obtain permission where required.

• Do not paste confidential, personal or restricted information into public AI tools.

• AI may hallucinate, misunderstand circuit polarity, give outdated specifications or produce biased wording.

• Human review and real experimental verification are mandatory.

16. Conclusion

The workflow shows how AI can reduce repetitive documentation effort and improve structure, readability and consistency. Its value depends on the quality of the student's source notes and the discipline of technical verification. Final time-saving and quality claims must be based on the completed measurement and comparison tables in this report.

Appendix A: Submission checklist

• Draft and final report (this report completed with actual data)

• README and user guide

• 5–7 slide presentation

• Before/after time and quality comparison

• Error log

• Automation script and sample notes

• Prompt library

• Verification report and evidence

• Before/after analysis

• Final demonstration

Appendix B: Student completion note

Before submission, replace all blanks and [TO BE PROVIDED] markers, insert your own circuit diagram and photographs, add actual references, record the four or more AI tools genuinely used, and obtain your instructor's approval for the selected circuit.